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Thesis Defense

Predicting SME Bankruptcy in France

A Machine Learning approach to Macro-Micro dynamics in a high-interest rate regime

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Context of the study

- 57,729 business failures in 2023.

→ **+35.8% increase** vs 2022.

- Shift from **accommodative** → **restrictive** monetary regime/policy (ECB rate hikes).
- **Objective**: Bridging the gap between firm-level accounting and macroeconomic shocks.

Problem statement & Research questions

Traditional models are often "macro-blind," failing to predict defaults during structural economic shifts.

- **RQ1**: To what extent does the integration of macroeconomic interest rate shifts improve the predictive accuracy of SME bankruptcy models compared to traditional frameworks relying solely on micro-level accounting indicators?
- **RQ2**: How does the non-linear interaction between firm-level leverage and monetary tightening act as a primary "risk activator" for corporate failure during a structural economic regime shift?
- **RQ3**: To what degree do advanced ML models maintain predictive robustness across structural breaks, and how does hyperparameter over-optimization affect their generalizability in volatile economic environments?

Literature review & Theoretical framework

01

Economic backbone

SMEs constitute the majority of French enterprises, facing inherent "information asymmetry" & credit rationing (Stiglitz & Weiss, 1981).

02

Evolution of models

Transitioning from static, accounting-only ratios (Altman, 1968; Ohlson, 1980) toward dynamic, macro-integrated frameworks.

03

The macro-determinant

Empirical evidence suggests failure rates are a function of monetary policy shifts, inflation, and GDP volatility (Liu, 2004; Tinoco & Wilson, 2013).

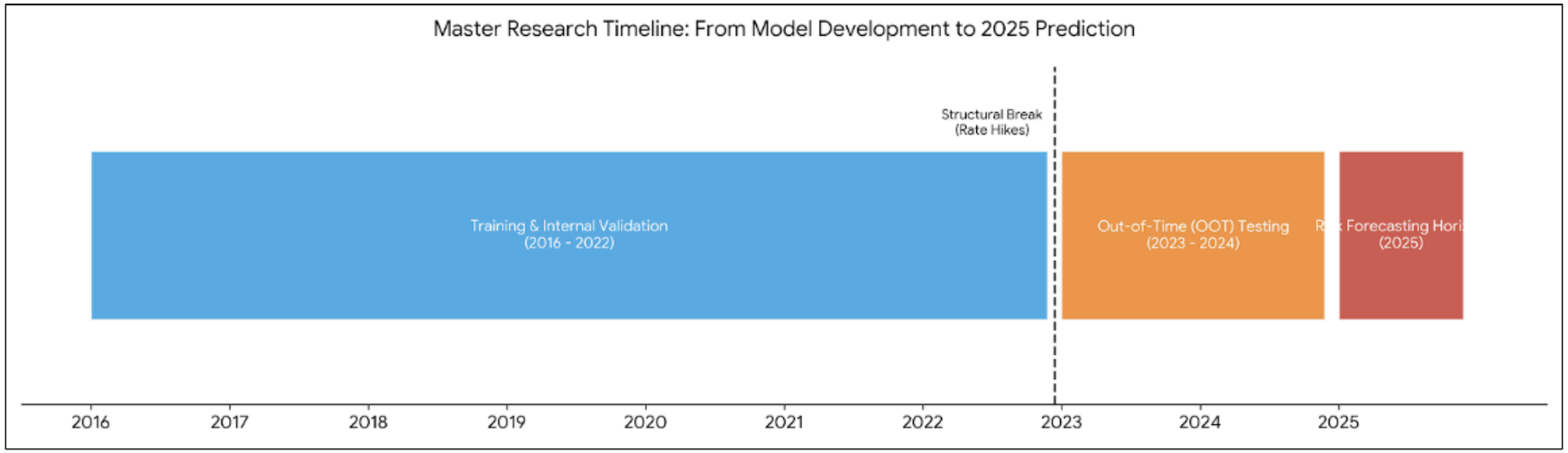
04

The interpretability frontier

Utilizing SHAP (Lundberg & Lee, 2017) to bridge the gap between "Black Box" machine learning and transparent, mechanistic risk insights.

Research Methodology

Methodological framework & Research design



Approach: Quantitative, longitudinal study using Supervised Machine Learning.

Objective: Developing a predictive diagnostic for SME insolvency under a restrictive monetary regime.

Target variable: Binary classification (Legal status: active vs. bankrupt).

Validation strategy: Out-of-Time (OOT) validation using 2023-2024 data.

Research Methodology

Data Sourcing & Filtering

- Primary source: Orbis (Bureau van Dijk) – ~268,000 French SMEs.
- Operational filters:
  - Firm size: **Total assets < €43 Million** (EU SME definition).
  - Temporal integrity: Requirement of **3 consecutive years** of financial data (T, T-1, T-2).
  - Observation baseline: Ensuring data availability for the 2016–2022 training window.

Research Methodology

Predictor variables

Category	*Variable*	*Formula*	*Academic support*
Liquidity	Current ratio (Current_Ratio_Lag1)	Current assets / Current liabilities	Altman (1968), Ohlson (1980)
Profitability	ROA (ROA_Lag1)	Net Income / Total Assets (Year T-1)	Altman (1968), Ohlson (1980)
	ROA_Trend	ROA (T-1) – ROA (T-2)	Barboza et al. (2017)
Leverage	Debt ratio (Debt_Ratio_Lag1)	Total Liabilities / Total Assets	Hernandez Tinoco & Wilson (2013)
	Interest burden (Interest_Burden_La	Interest Expense / EBITDA	Gupta & Gregoriou (2018)
Scale	Total assets (Total_Assets_Lag1)	Natural log of total assets	Ohlson (1980)
	Revenue_Growth	(Sales (T-1) – Sales (T-2)) / Sales (T-2)	Jones (2017)
Operational	Maturity_Score	Composite score (Firm Age & Stability)	Gupta & Gregoriou (2018)

Category	*Variable*	*Formula*	*Academic support*
Macroeconomic	Macro_GDP_Growth	Annual GDP Growth Rate (France)	Liu (2004)
	Macro_Inflation	HICP Inflation Rate	Liu (2004), Jones (2017)
	Macro_Interest_Rate	ECB Marginal Lending Facility Rate	Proposed in this study
	Macro_Unemployment	National Unemployment Rate	Liu (2004)
Interaction	Debt_Interest_Interact	Debt_Ratio_Lag1 × Macro_Interest_Rate	Proposed in this study

Research Methodology

Feature engineering

- The new variable: Debt_Interest_Interact.
- Formula: Debt_Ratio (T-1) × Macro_Interest_Rate (T).
- Rationale: Accounting ratios are static; this variable "activates" leverage risk based on the real-time cost of capital.
- Preprocessing: Addressing class imbalance via optimized class weighting and feature scaling.

Research Methodology

Technical implementation: Algorithm selection

- Linear benchmark: Logistic Regression for baseline interpretability.
- Non-Linear models: XGBoost & LightGBM.
- Scientific justification:
  - Capability to capture high-dimensional non-linear relationships.
  - Robustness to multicollinearity between macro and micro features.
  - Internal handling of missing values without biased imputation.

Research Methodology

Technical validation – TimeSeriesSplit mechanism

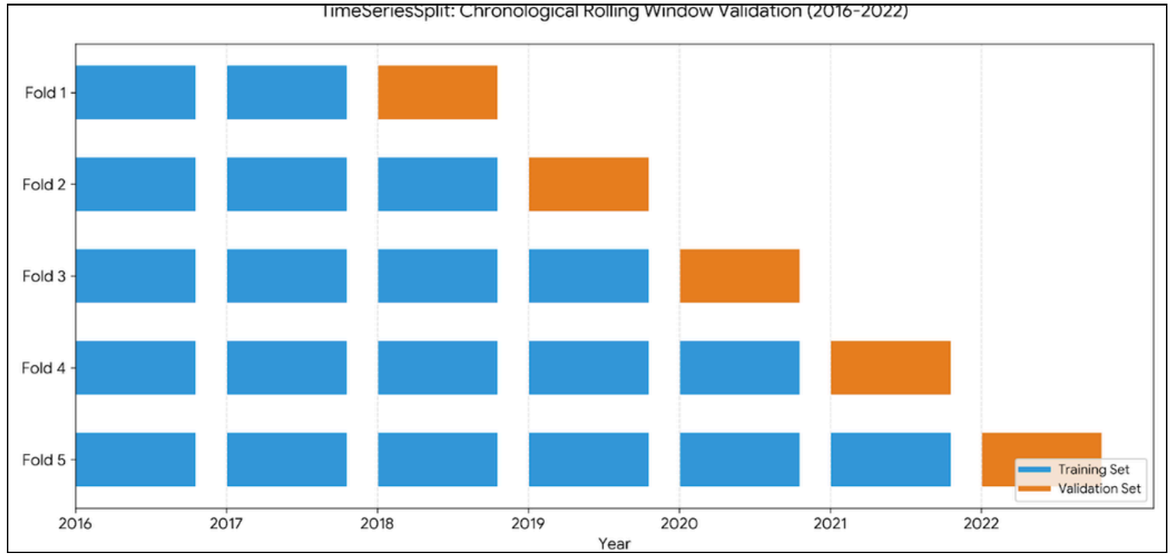
- Traditional K-Fold Cross-Validation (shuffling) is unsuitable for longitudinal financial data as it leads to Temporal Data Leakage.

→ Implementation of TimeSeriesSplit to respect the chronological order of SME defaults.

- Mechanism Workflow:
  - Fold 1: Train on [T1], Validate on [T2].
  - Fold 2: Train on [T1+T2], Validate on [T3].
  - The training set expands chronologically, ensuring the model always predicts the future using only the past.

→ Eliminates look-ahead bias, preventing the model from using future information to predict past events.

→ Ensures stability of the "Macro-Micro" interaction weights across different economic sub-periods.



Results & Discussion

01 The best model

Baseline XGBoost.

02 Comparison

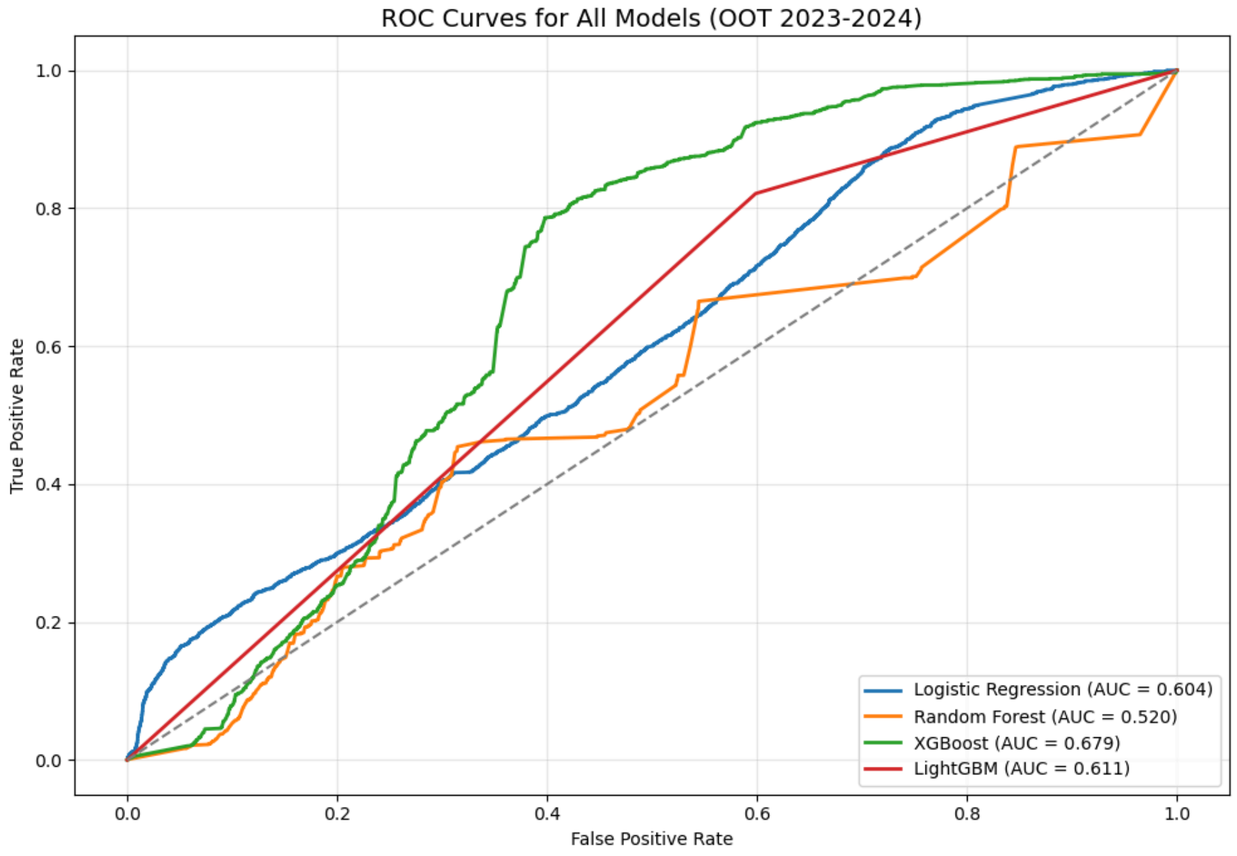
- XGBoost AUC (OOT): 0.68 (Superior stability).
- Logistic Regression AUC: 0.60.

03 Inference

Non-linear algorithms are significantly better at navigating the "regime change" of 2023.

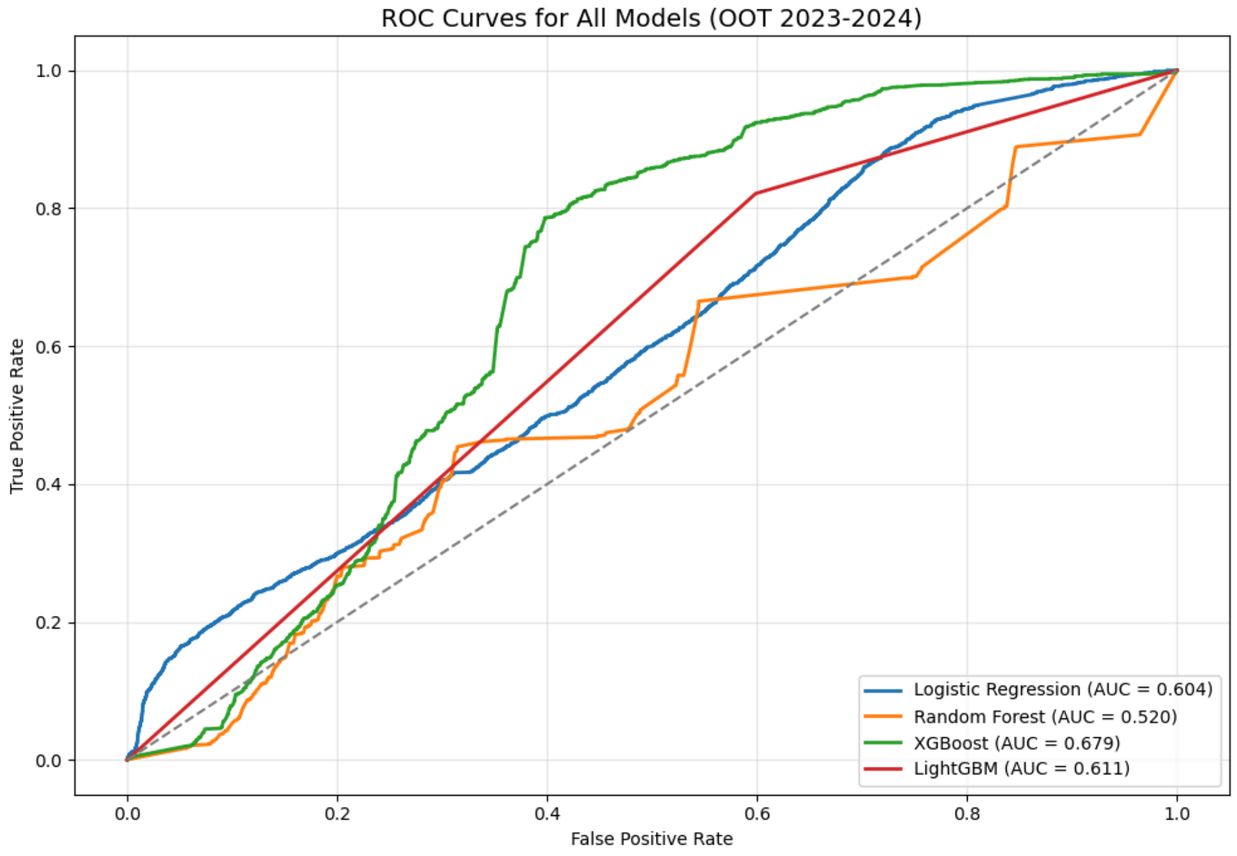
	AUC	***Precision***	***Recall***	***F1-score***
Logistic Regression	0.6	0.01	0.98	0.01
Baseline XGBoost	0.68	0	0.02	0
Tuned XGBoost	0.57	0.01	0.13	0.01

In-Depth ROC Analysis



- Performance Comparison: Baseline XGBoost (AUC 0.68) maintains a consistent curve above the diagonal, while Tuned XGBoost (AUC 0.57) exhibits a flatter trajectory, tiptoeing near the random line.
- The "Sub-Diagonal" Phenomenon: Identifying segments where the ROC curve falls below the 45-degree diagonal line (AUC < 0.5).

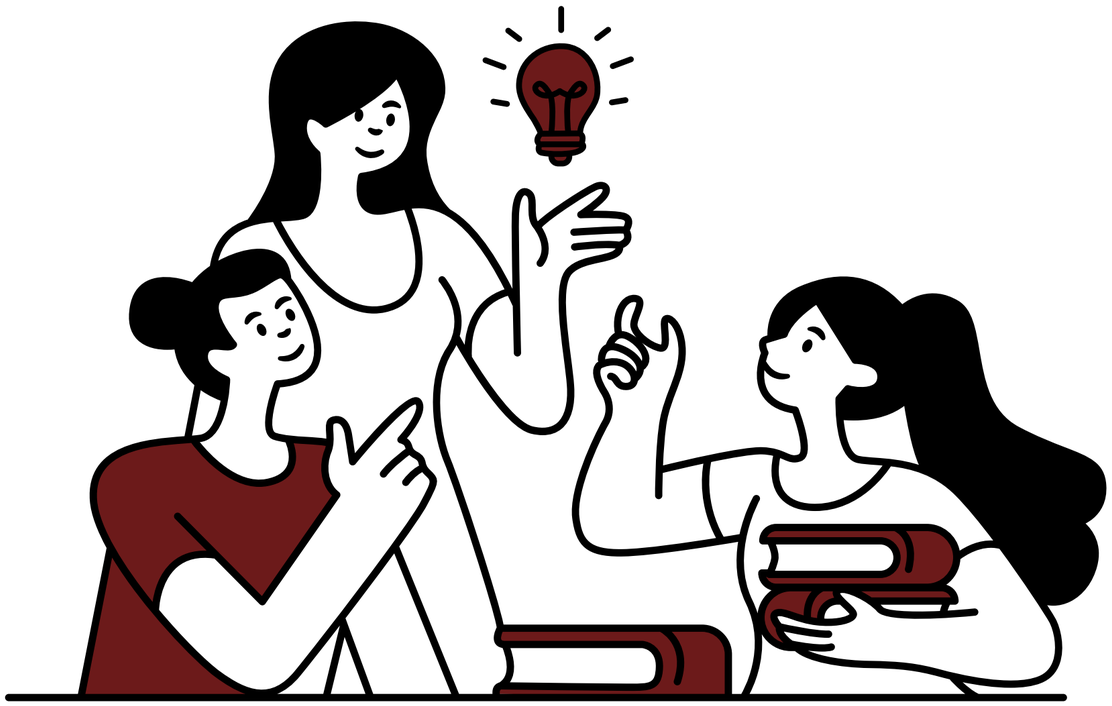
In-Depth ROC Analysis



- Hypothesis:
  - During the 2016–2022 training regime, the model learned that high leverage was a proxy for "growth and borrowing capacity."
  - The interest rate shock inverted this relationship—high leverage transformed from a sign of strength into a "death sentence."
  - The tuned model was "over-fitted" to the old data; it confidently predicted survival for firms that were actually the most vulnerable under the new regime.

The "Tuning Paradox"

- The performance collapse: Tuned XGBoost AUC fell to 0.57.
- Root cause: "Regime-trapping." Hyperparameters **optimized for the 2016-2022** accommodative period **failed to generalize to the 2023** shock.
- During structural breaks, algorithmic robustness outweighs historical precision.



SHAP feature hierarchy

Top 3 predictors:

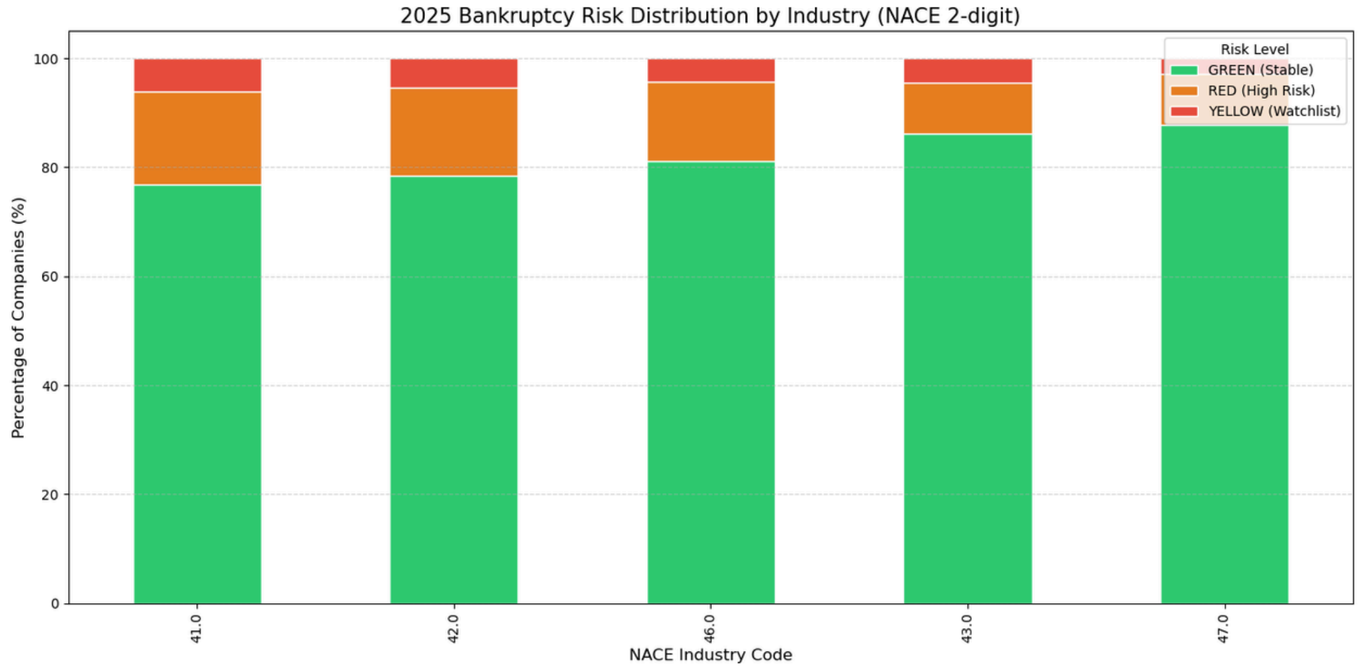
- **Total_Assets_Lag1**: Firm scale remains the primary anchor of resilience.
- **Debt_Ratio_Lag1**: The most critical micro-vulnerability.
- **Debt_Interest_Interact**: The innovation variable - proving that the interaction between debt and interest rates is a top-3 determinant of failure.

Structural Divergence & Bankruptcy Critical Thresholds

Financial Indicator	*Group 0*	*Group 1*	*Performance Divergence*
Return on Assets (ROA)	0.05	-0.03	Profit-to-Loss Shift
Total Assets (EUR)	847,217.00	98,485.00	8.6x Scale disparity
Debt Ratio	0.56	0.73	Elevated leverage

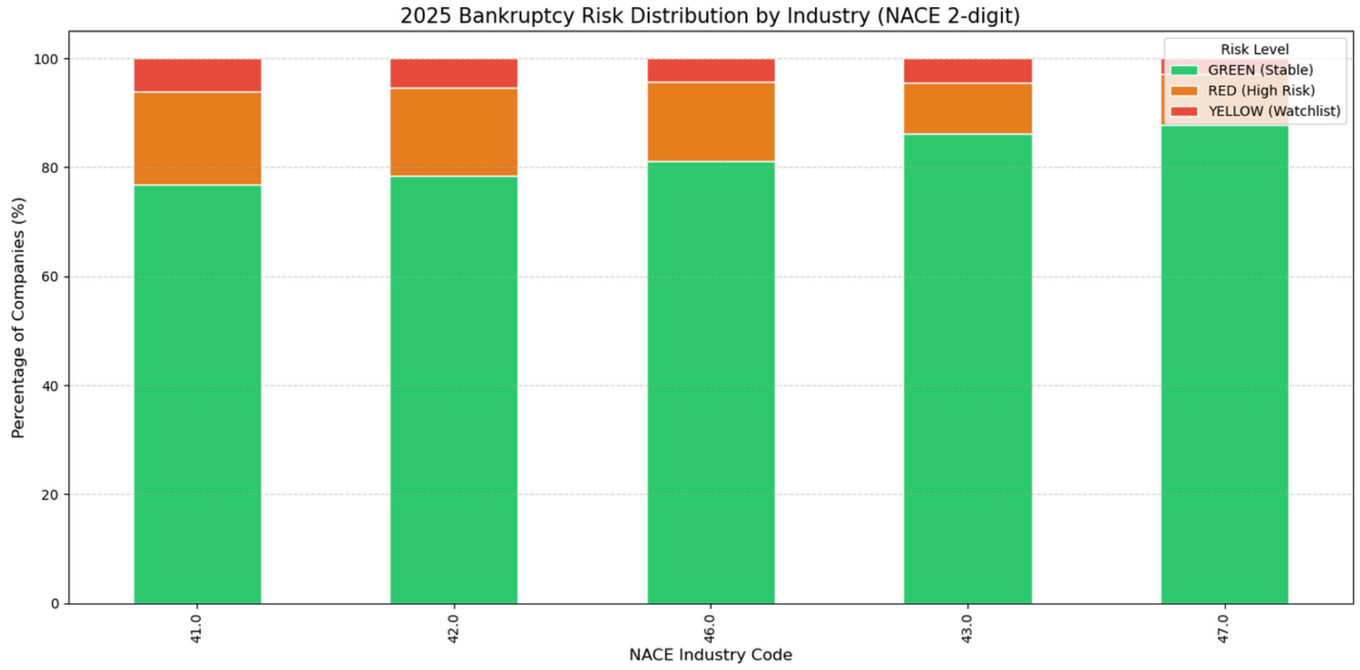
- Survivors maintain a median ROA of 0.05, providing a robust internal liquidity buffer.
- Bankrupt entities exhibit a −0.03 deficit, representing a total exhaustion of operating margins to service financial obligations.
- A median Debt Ratio of 0.73 serves as a structural vulnerability threshold.
- Under restrictive monetary conditions, this level of leverage transitions from a financing tool to a primary solvency threat.
- The intersection of negative operating margins and elevated leverage levels creates a systemic insolvency risk.
- Inference: Firms with a −3% ROA lack the mathematical capacity to cover the rising cost of capital (interest rates ≥3.5%).

NACE-Specific Vulnerabilities



- Construction (NACE 41): Exhibits the highest probability of default, driven by high capital intensity and sensitivity to mortgage rate-driven demand contraction.
- Wholesale trade (NACE 46): Significant vulnerability due to thin operating margins and heavy reliance on short-term working capital financing.

NACE-Specific Vulnerabilities



- Leverage sensitivity: Sectors with high fixed costs (Construction) face immediate debt-service pressure as the cost of capital exceeds structural thresholds.
- Margin compression: Sectors with low pricing power (Wholesale) are unable to pass increased borrowing costs to consumers, leading to rapid liquidity attrition.

Exposure at risk

Risk level	*Company*	*Total debt at risk (EUR)*	*Count share*	*Debt share*
GREEN (Stable)	153757	156,184,208,575.17	84.95%	74.29%
RED (High Risk)	20178	39,787,178,835.53	11.15%	18.93%
YELLOW (Watchlist)	7067	14,254,489,714.69	3.90%	6.78%

The **high-risk** cohort constitutes **11.15%** of the total SME sample.

This **minority group** accounts for a disproportionate **18.93%** of the total systemic **debt**.

Total estimated **debt-at-risk** in the restrictive regime: **€39.79 Billion**.

The €39.79B exposure is a conservative lower bound, as it excludes the survival bias inherent in the 3-year data reporting requirement.

Conclusion

- Integrating macroeconomic indicators significantly enhances model stability during structural shifts, achieving a stable OOT **AUC of 0.68**.
- **Scale (Total assets)** and **leverage (Debt ratio)** remain fundamental; however, the Debt-Interest Interaction has emerged as a top-3 failure determinant.
- **Over-optimization** on historical data from accommodative monetary regimes (2016–2022) results in catastrophic performance degradation (**AUC 0.57**) during restrictive cycles.
- Identified a high-risk concentration where **11.15% of firms** account for **€39.79 Billion** in systemic debt-at-risk.

Managerial and Macro-Prudential Implications

- **For financial institutions:**
  - Transition from static, backward-looking accounting models (IFRS 9) toward forward-looking, macro-dynamic frameworks.
  - Implementation of Conditional Stress Testing that accounts for the non-linear relationship between interest rates and corporate leverage.
- **For economic policymakers:**
  - Prioritize targeted liquidity interventions for Construction (NACE 41) and Wholesale (NACE 46) to mitigate sectoral contagion.
  - Mitigation of systemic credit supply contraction by monitoring the inverse relationship between capital costs and SME solvency.

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Thank You

Thank you for your attention